



University of Colombo School of Computing
SCS 1301 - Data Structures & Program Design in C
Lab Sheet 02

Instructions for Students (Ubuntu Environment)

1. This practical is to be completed using the **Ubuntu Operating System** (Terminal).
2. Open the Terminal using one of the following methods:
 - Press **Ctrl + Alt + T**
 - Or search for “**Terminal**” from the applications menu
3. All commands must be typed **exactly as given**.
 - Linux is **case-sensitive** (e.g., **Desktop** is different from **desktop**).
4. Execute each command by pressing the **Enter** key.
5. Carefully **observe the output** displayed in the Terminal.
6. In **Part 01**, follow the commands in the given order.
 - Do not skip steps, as each command depends on the previous one.
7. In **Part 02 (Dialogue Section)**:
 - Read each conversation carefully
 - Identify the correct Linux command
 - Write both:
 - **command**
 - **expected output** (Attach Screenshot)
8. If you make a mistake:
 - Use the **Up Arrow** key to recall previous commands

9. Submission - Create a PDF with output screenshots and Commands.

PART - A

1. **pwd** - Displays the current working directory.
2. **ls** - Displays the current working directory.
3. **ls -l** - Displays detailed information (permissions, owner, size, date).
4. **ls -a** - Includes hidden files (starting with **.**).
5. **cd**
 - a. **cd ~** - Moves to the home directory.
 - b. **cd Desktop** - Changes directory to Desktop.
6. **mkdir**
 - a. **mkdir test_folder** - Creates a folder named test_folder.
7. **touch**
 - a. **touch file1.txt** - Creates an empty text file.
8. **echo**
 - a. **echo "Hello Linux" > file1.txt** - Writes text into the file.

9. **cat**
 - a. **cat file1.txt** - Displays file contents.
10. **cp**
 - a. **cp file1.txt file2.txt** - Creates a copy.
11. **mv**
 - a. **mv file2.txt renamed.txt** - Renames the file.
12. **rm**
 - a. **rm renamed.txt** - Removes the file.
13. **cd ..** - Moves to the parent directory.
14. **clear** - Clear the terminal
15. **whoami** - Displays logged-in username
16. **date** - Displays system date & time
17. **cal** - Displays current month calendar
18. **history** - Shows previously used commands
19. **find . -name**
 - a. **find . -name "file1.txt"** - Searches file in current directory
20. **ls > list.txt** - Saves output into a file
21. **ls >> list.txt** - Adds output without overwriting

PART - B (Dialogue Section)

Dialogue 01:

Person 01: "Where are we right now in the system?"

Person 02: "Let me check."

Dialogue 02:

Person 01: "Can you show me what files are here?"

Person 02: "Sure."

Dialogue 03:

Person 01: "Where is the Desktop folder? Can you go there?"

Person 02: "Okay, navigating now."

Dialogue 04:

Person 01: "Create a folder called project_work."

Person 02: "Done."

Dialogue 05:

Person 01: "Go inside that folder."

Person 02: "Entering now."

Dialogue 06:

Person 01: "Create a file named **notes.txt**."

Person 02: "File created."

Dialogue 07:

Person 01: "Write 'Linux is powerful' into the file."

Person 02: "Added."

Dialogue 08:

Person 01: "Show me what's inside the file."

Person 02: "Here it is."

Dialogue 09:

Person 01: "Make a copy of this file."

Person 02: "Copy created."

Dialogue 10:

Person 01: "Delete the copied file."

Person 02: "Removed."

PART - C (Dialogue Section)

Dialogue 01

Person 01: "I think there are some hidden files here. Can you show all of them?"

Person 02: "Yes, listing everything, including hidden files."

Dialogue 02:

Person 01: "Create a folder called **project** inside Desktop and go into it."

Person 02: "Okay, creating and entering now."

Dialogue 03:

Person 01: "I need three files: **a.txt**, **b.txt**, and **c.txt**."

Person 02: "I'll create them together."

Dialogue 04:

Person 01: "I want to see full details of all files in this folder."

Person 02: "Displaying detailed list."

Dialogue 05:

Person 01: "Find the word 'apple' even if it's in uppercase or lowercase."

Person 02: "I'll perform a case-insensitive search."

Dialogue 06:

Person 01: "How many lines are there in this file?"

Person 02: "I'll count them."

Dialogue 07:

Person 01: "Make this file read-only."

Person 02: "I'll remove write permission."

Dialogue 08:

Person 01: "I want to make this file executable."

Person 02: "Adding execute permission."

Dialogue 09:

Person 01: "Count how many lines contain the word 'banana'."

Person 02: "I'll combine commands."

Dialogue 10:

Person 01: "Remove duplicate lines and show only unique ones."

Person 02: "Processing now."

Dialogue 11:

Person 01: "Show lines that do NOT contain the word 'apple'."

Person 02: "Filtering them out."

Dialogue 12:

Person 01: "Sort this file in reverse alphabetical order."

Person 02: "Sorting now."

-End-